

Please ensure this tracking sheet is completed as a group, as it will be used to monitor your individual participation within the team.

Team Members	Role/ Responsibility	Email					
Ivan Ineza Hakizimana	API & Security Developer — wrote api/server.py: all 5 CRUD endpoint handlers, the Basic Auth check, and the input-validation logic (check_data function: required fields, type checks, negative-amount rejection)	i.inezahak@alustudent.com					
Hezekiah Samuel Epodoi	Data Parsing & JSON Modeling — wrote dsa/parse_xml.py: the regex extraction logic for each SMS message type, the transaction schema (id, type, amount, sender, receiver, balance, transaction_id, timestamp), and the XML-to-JSON conversion	h.epodoi@alustudent.com					
Muhozi ntwali Devis	DSA & Algorithms — wrote dsa/search_compare.py: implemented linear search, dictionary lookup, and binary search from scratch, plus the timing/benchmark code that measures and compares them	d.muhozin@alustudent.com					
Task Allocation and Tracker							
	Task Allocated	Assigned Member	Deadline	Completion Status (Not Started/In Progress/Completed/Did not participate)	Reviewed by Team? (Yes/No)	Comments	
	Write regex parsing rules for each SMS type in parse_xml.py	Hezekiah Samuel Epodoi	28/09/2026	Completed	Yes	Handles 12 transaction types, 1691 records	
	Design the transaction JSON schema (fields, types)	Hezekiah Samuel Epodoi	28/09/2026	Completed	Yes	id, type, amount, sender, receiver, balance, transaction_id, timestamp	
	Write GET/POST/PUT/DELETE handlers in server.py	Ivan Ineza Hakizimana	28/09/2026	Completed	Yes	Full CRUD, routes by regex-free path splitting	
	Write Basic Auth check and check_data input validation	Ivan Ineza Hakizimana	28/09/2026	Completed	Yes	Returns 401 / 400 correctly	
	Implement linear search function	Muhozi ntwali Devis	28/09/2026	Completed	Yes	O(n) scan of the transaction list	
	Implement dictionary lookup and binary search functions	Muhozi ntwali Devis	28/09/2026	Completed	Yes	O(1) and O(log n) respectively	
	Write benchmark/timing code and compare results	Muhozi ntwali Devis	28/09/2026	Completed	Yes	Dictionary ~300x faster at 1691 records	
	Test all endpoints with curl, verify status codes	Ivan Ineza Hakizimana	28/09/2026	Completed	Yes	9 test cases, all passing	
	Debug and cross-check parser output against raw XML	Hezekiah Samuel Epodoi	28/09/2026	Completed	Yes	Confirmed 0 unparsed ("other") records	
	Assemble README/docs/report (shared, not graded individually)	All three of us	29/09/2026	Completed	Yes	Non-code deliverable, done as a group	
Meeting Attendance Log							
Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps		
Friday, September 25.	Kickoff: split coding work three ways	Hezekiah Samuel Epodoi	All 3 of us	Agreed each member owns one script (parsing, API, DSA) so grading can trace code to person; agreed on the transaction schema fields together	Each member starts coding their script independently		
Sunday, September 27.	Schema alignment check	Hezekiah Samuel Epodoi	All 3 of us	Confirmed the field names/types coming out of parse_xml.py matched what server.py and search_compare.py expected	Ivan and Devis adjust their scripts to the agreed schema		
Monday, September 28.	Integration	Muhozi ntwali Devis	All 3 of us	Ran all three scripts together, tested API endpoints live, ran the DSA benchmark against the real parsed data	Fix any bugs found during integration testing		
Tuesday, September 29.	Final code review	Ivan Ineza Hakizimana	All 3 of us	Each member walked the other two through their own script line by line so everyone could verify individual contributions	Submit		